

NMAP

COMPLETE GUIDE FOR BLUE TEAMERS

The SOC Analyst's Handbook for Network Discovery, Scanning, and Security Assessment

Discover Hosts → Scan Ports → Detect Services → Assess Security
Host Discovery → Port Scanning → Service Detection → OS Detection → NSE Scripts → Risk Assessment

See More. Know More. Defend Better.

Made for SOC Analysts | Discover · Scan · Analyze · Defend

Your journey to stronger defenses starts here!

Text extracted from the original PDF guide (blueteamers.io).

NMAP – Network Mapper

"The Most Popular Network Scanning Tool in Cybersecurity"

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What Is Nmap?

Nmap (Network Mapper) is an open-source tool used to discover devices, identify open ports, detect running services, and gather information about systems on a network.

It helps security professionals understand what is connected to a network and how those devices can communicate.

In the guide's diagram, Nmap sits at the centre and scans network devices: a server, an employee laptop, a printer, an IP camera and a router.

Uses of Nmap

- **Host Discovery** – Find active devices on a network.
- **Port Scanning** – Identify open communication ports.
- **Service Detection** – Detect services running on a system.
- **Operating System Detection** – Identify Windows, Linux, or other operating systems.
- **Security Assessment** – Discover potential security weaknesses.

Real-World Example

Scenario: A company has 100 devices connected to its office network. The security team wants to know:

- Which devices are active?
- Which ports are open?
- Which services are running?

Instead of checking each device manually, they use Nmap to scan the entire network and collect this information within minutes.

Screenshot (Kali Linux) – Verifying Nmap Installation

```
(kali@kali) - [~]
$ nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.9 nmap-libssh2-1.10.0
Nmap script engine: LuaJIT 2.1.0-beta3
Nmap Scripting Engine Version: 3.0
```

Notes

- Created by Fyodor (Gordon Lyon)
- Free and Open Source
- Works on Windows, Linux, and macOS
- Widely used by SOC Analysts and Penetration Testers
- One of the most important cybersecurity tools

"Before finding vulnerabilities, you must first discover what exists on the network — that's where Nmap begins."

Installing Nmap

Nmap is available for Windows, Linux and macOS. Follow the steps below to install Nmap.

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Installation Overview

1. Download Nmap
2. Install Nmap
3. Verify Installation
4. Ready to Scan!

Windows Installation

1. Visit <https://nmap.org/download.html>
2. Download the latest Windows installer.
3. Run the installer (.exe file).
4. Follow the setup wizard.
5. Open Command Prompt.
6. Verify Nmap installation.

Linux Installation (Kali / Ubuntu)

Open terminal and run the following commands:

```
$ sudo apt update
$ sudo apt install nmap -y
```

(Kali Linux already has Nmap pre-installed)

Verify Installation

Check Nmap version:

```
$ nmap --version
```

If you see version information like in the screenshot, Nmap is installed successfully!

Screenshot (Kali Linux) – Verifying Nmap Installation

```
(kali@kali)-[~]
$ nmap --version
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.9 nmap-libssh2-1.10.0
Nmap script engine: LuaJIT 2.1.0-beta3
Nmap Scripting Engine Version: 3.0
```

Notes

- Nmap requires administrator / root privileges for advanced scans.
- Kali Linux comes with Nmap pre-installed.
- Always use the latest version for better features and security.
- After installation, you are ready to start scanning networks!

Understanding IP Addresses

Every device connected to a network has a unique IP address. Nmap uses IP addresses to scan and discover devices.

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What Is an IP Address?

An IP (Internet Protocol) address is a unique identifier assigned to each device on a network so they can communicate with each other. The most common version is IPv4, which consists of 4 numbers (0–255) separated by dots.

Example: 192.168.1.10

Example Network

The Internet connects to a router (192.168.1.1), which connects to:

Device	IP Address
Router	192.168.1.1
Laptop	192.168.1.5
Desktop	192.168.1.8
Printer	192.168.1.15
Server	192.168.1.10

Common Private IP Ranges

Range	Description
192.0.0.0 – 10.255.255.255	Class A Private
172.16.0.0 – 172.31.255.255	Class B Private
192.168.0.0 – 192.168.255.255	Class C Private

Important Notes

- Private IP addresses are used inside local networks (home, office, enterprise).
- Public IP addresses are used on the internet to identify networks globally.
- Nmap can scan both private and public IP addresses.

Real-World Example

Scenario: A system administrator suspects an unknown device is connected to the office network. He uses Nmap to scan the network range 192.168.1.0/24 to find all active devices and their IP addresses.

Goal: Discover, Identify, and Monitor. Stay Secure.

Check Your IP (Kali Linux)

```
(kali@kali) - [~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    qlen 1000
    link/ether 08:00:27:3c:5a:1b brd ff:ff:ff:ff:ff:ff
```

```
inet 192.168.1.25/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
    valid_lft 86386sec preferred_lft 86386sec
inet6 fe80::a00:27ff:fe3c:5a1b/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
```

The highlighted line (`inet 192.168.1.25/24`) is the machine's IP address on the network.

Notes

- IP addressing is the foundation of networking.
- Before scanning, you must know the target IP or network range.
- Most internal investigations start with discovering live hosts.
- Nmap helps you see the “big picture” of the network.

Understanding IP addresses is the first step to mastering network reconnaissance.

Host Discovery (Ping Scan)

Host discovery helps us find active devices on the network. It does not scan ports, it only checks which hosts are alive.

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Command

```
nmap -sn 192.168.1.0/24
```

Breakdown

- `-sn` : Ping Scan (No Port Scan)
- `192.168.1.0/24` : Target Network

What Does It Do?

- Sends packets to hosts in the network.
- Finds which hosts are up/active.
- Does NOT scan any ports.
- Shows IP and MAC address of active devices.

How It Works

The attacker machine (Kali Linux) sends requests (ICMP Echo Requests) to each host:

Device	IP Address	Status
Router	192.168.1.1	UP
Desktop	192.168.1.5	UP
Printer	192.168.1.10	UP
Mobile	192.168.1.20	DOWN

If the host replies, Nmap marks it as UP. If no reply, it is marked as DOWN.

Example Output (Kali Linux)

```
(kali@kali)-[~]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 10:30 IST
Nmap scan report for 192.168.1.1
Host is up (0.0010s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.1.5
Host is up (0.00078s latency).
MAC Address: 00:50:56:C0:00:05 (VMware)

Nmap scan report for 192.168.1.10
Host is up (0.00062s latency).
MAC Address: 00:0C:29:1A:2B:3C (VMware)

Nmap done: 256 IP addresses (3 hosts up) scanned in 2.35 seconds
```

Output annotations: 192.168.1.1 – Router is alive. 192.168.1.5 – Desktop is alive. 192.168.1.10 – Printer is alive. 3 hosts are up on this network.

Real-World Example

A SOC Analyst wants to know which devices are currently active in the office network. He runs a ping scan to quickly list all live systems before performing detailed scans.

Notes

- Fast and safe scan.
- Useful during reconnaissance.
- Helps to identify targets for further scanning.
- Works even if firewall blocks port scans.
- Commonly used by Blue Teamers for asset discovery.

Quick Tip

Always start your assessment with a host discovery scan! It helps you understand the network before deep scanning.

Ports 101

Understanding Network Ports – Ports are like doors in a building. They allow data to flow in and out of a system.

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What Is a Port?

A port is a logical endpoint that enables communication between devices. Each port is assigned a number (0 – 65535).

Example: target system 192.168.1.10 with open ports (doors): 22 (SSH), 80 (HTTP), 443 (HTTPS), 3306 (MySQL).

Common Ports

Port	Service	Description
21	FTP	File Transfer Protocol
22	SSH	Secure Shell
23	TELNET	Remote Terminal
53	DNS	Domain Name System
80	HTTP	Web Traffic (Unsecured)
443	HTTPS	Web Traffic (Encrypted)
3389	RDP	Remote Desktop Protocol

Real-World Example

Your browser connects to a web server (192.168.1.10) on port 80 (HTTP) or port 443 (HTTPS). If port 80 is open, the website loads. If port 80 is closed, you cannot connect.

Basic Port Scan

Command to scan common ports:

```
nmap 192.168.1.10
```

Screenshot (Kali Linux)

```
(kali@kali)-[~]
$ nmap 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 11:15 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
3306/tcp  open  mysql

Nmap done: 1 IP address (1 host up) scanned in 1.24 seconds
```

Nmap found 4 open ports on the target system.

Notes

- Not all open ports are dangerous.
- Always investigate the service running behind the port.

- Attackers look for unnecessary open ports.
- Close unused ports for better security.

Key Takeaway

Understanding ports is the foundation of network scanning and security assessment.

Port Scanning

Scanning for Open Doors – Port scanning is the process of checking a target system to find open, closed, or filtered ports.

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What Is Port Scanning?

Port scanning helps us identify which ports on a target system are:

- **OPEN** (Accepting connections)
- **CLOSED** (Not accepting connections)
- **FILTERED** (Blocked by firewall / not reachable)

Knowing open ports helps attackers or defenders understand what services are exposed.

How It Works

The scanner (Kali Linux) sends a SYN packet across the network to the target system (192.168.1.10). The target answers with SYN/ACK (open) or RST (closed). Nmap sends packets to the target ports and analyzes the responses to determine their status.

Common Port States

State	Meaning
OPEN	Port is open and the service is accepting connections.
CLOSED	Port is closed and the service is not accepting connections.
FILTERED	Port is blocked by firewall or cannot be determined.

Real-World Example

A company web server (192.168.1.10) is running. A security analyst scans it to see which ports are open. If port 80 (HTTP) and 443 (HTTPS) are open, users can access the website. If port 22 (SSH) is open, someone may try to log in remotely.

Basic Port Scan

This is a simple scan that checks the most common 1000 ports.

```
nmap 192.168.1.10
```

Screenshot (Kali Linux)

```
(kali@kali)-[~]
$ nmap 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 11:30 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
3306/tcp   open  mysql

Nmap done: 1 IP address (1 host up) scanned in 1.24 seconds
```

Explanation of Output

- Port 22 is open (SSH)

- Port 80 is open (HTTP)
- Port 443 is open (HTTPS)
- Port 3306 is open (MySQL)
- These services are running on the target.

Tips

- More open ports = more attack surface.
- Regularly scan your own systems.
- Close unused ports to reduce risk.

Notes

- Default scan checks top 1000 ports.
- Use `-p` to scan specific ports.
- Use `-p-` to scan all 65535 ports.
- Port scanning is the first step in any assessment.

Service Version Detection

(What services are running?) – Service version detection helps identify the exact versions of services running on open ports. This information is useful for security analysis.

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Command

```
nmap -sV 192.168.1.10
```

Breakdown: `-sV` : Enables service version detection. `192.168.1.10` : Target IP address.

How It Works

Nmap (Kali Linux) sends probes to the target (192.168.1.10). The target returns service responses, and Nmap detects the service and version.

Common Services Nmap Detects

Port	Service	Example Version
22/tcp	SSH	OpenSSH 8.2p1 Ubuntu
80/tcp	HTTP	Apache httpd 2.4.41
443/tcp	HTTPS	nginx 1.18.0 (Ubuntu)
3306/tcp	MySQL	MySQL 5.7.33
21/tcp	FTP	vsftpd 3.0.3
8080/tcp	HTTP-Alt	Tomcat 9.0.41

Real-World Example

A SOC Analyst scans a web server to know which web server and SSH version is running.

Screenshot (Kali Linux)

```
(kali@kali)-[~]
$ nmap -sV 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 12:02 IST
Nmap scan report for 192.168.1.10
Host is up (0.00071s latency).

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.5
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
443/tcp   open  https    nginx 1.18.0 (Ubuntu)
3306/tcp   open  mysql    MySQL 5.7.33-0ubuntu0.18.04.1

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/
```

Output Explained

- 22/tcp (SSH) – OpenSSH version 8.2p1
- 80/tcp (HTTP) – Apache version 2.4.41
- 443/tcp (HTTPS) – nginx version 1.18.0
- 3306/tcp (MySQL) – MySQL version 5.7.33

Use Case

- Detect outdated software
- Find vulnerable versions
- Identify exposed services
- Plan next steps for assessment

Notes

- Version detection (`-sV`) takes more time.
- Some services may not return exact versions.
- Combine with other scans for better results.

OS Detection (-O)

Identify the Operating System of a Target – Nmap can guess the operating system of a remote host using TCP/IP fingerprinting.

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What Is OS Detection?

Nmap sends a series of TCP/IP packets to the target and analyzes the responses. From the responses, Nmap compares the behavior with its database (nmap-os-db) to guess the OS.

Command

```
sudo nmap -O 192.168.1.10
```

Run with sudo/administrator privileges for better accuracy.

How It Works (TCP/IP Fingerprinting)

1. Nmap sends specially crafted packets.
2. The target responds based on its OS behavior.
3. Nmap analyzes TCP/IP stack characteristics.
4. The OS is matched with the closest fingerprint.

Diagram: Target system (192.168.1.10) → Nmap analysis (TCP/IP fingerprinting) → Detected OS (example output: Windows 10, Ubuntu Linux, Windows Server 2019).

Windows vs Linux Detection

Characteristic	Windows	Linux
TTL value	Usually 128	Usually 64
Window size	Larger	Smaller
ICMP behavior	Different	Different
TCP options	Different order	Different order

Nmap uses many such characteristics to identify the OS.

Why SOC Analysts Need It

- Identify unknown devices on the network.
- Detect unauthorized operating systems.
- Prioritize patching for vulnerable OS.
- Improve situational awareness in investigations.
- Support incident response and threat hunting.

Real-World Example

An unknown device appears on the office network. The SOC analyst uses Nmap to determine:

- It is running Windows 11
- It is running Ubuntu Linux
- It is running Windows Server

This helps the team take the right actions.

Example Output (Kali Linux)

```
(kali@kali) - [~]
```

```
$ sudo nmap -O 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 12:15 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
Device type: general purpose
Running: Microsoft Windows 10|2016|2019
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows 10 1809 - 22H2, Microsoft Windows Server 2016
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.45 seconds
```

What Is Shown Here?

- Open ports: 22, 80, 443
- Device type: General Purpose
- OS Guess: Windows 10 / Windows Server 2016
- Accuracy depends on the target and privileges.

Notes

- OS detection (-O) requires root/administrator privileges.
- Accuracy improves when more ports are open.
- Firewalls may block some probes and affect results.

| *Knowing the OS is the first step towards securing the system!*

TCP SYN Scan (-sS)

The Most Popular Nmap Scan – TCP SYN Scan is a stealthy and efficient scan technique used by security professionals and blue teamers.

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What Is SYN Scan?

SYN Scan is a “half-open” scan. It sends a SYN packet to the target port and analyzes the response without completing the full TCP handshake.

Command

```
sudo nmap -sS 192.168.1.10
```

Run with sudo/administrator privileges for raw packet crafting.

Why Is It Stealthy?

- Does not complete the TCP handshake.
- Leaves minimal logs on the target system.
- Harder to detect by intrusion detection systems.
- Fast and efficient.

When to Use?

- Internal network assessments
- Firewall testing
- Penetration testing
- SOC investigations
- Threat hunting

Real-World Example

A SOC analyst wants to stealthily check which ports are open on a critical server without triggering alerts. SYN Scan helps them discover exposed services while remaining low-profile.

TCP Handshake Explained

1. **SYN** – The scanner sends a SYN packet.
2. **SYN/ACK** – If the port is open, the target replies with SYN/ACK.
3. **RST** – The scanner sends RST to close the connection.

Interpretation

Result	Meaning
OPEN	Received SYN/ACK response.
CLOSED	Received RST response.
FILTERED	No response or ICMP error (blocked).

Kali Linux Output (Deep Explanation)

```
(kali@kali)-[~]
$ sudo nmap -sS 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 12:35 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).
Not shown: 997 closed tcp ports (reset)
```

```
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.5
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
443/tcp   open  https    nginx 1.18.0 (Ubuntu)
MAC Address: 00:0C:29:1A:2B:3C (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.24 seconds
```

Deep Output Explanation

1. Command used: SYN Scan on 192.168.1.10
2. Nmap version and scan start time
3. Target scanned: 192.168.1.10
4. Host is UP and responded in 0.00078s
5. 997 ports are closed (RST received)
6. Open ports found with their services
7. MAC address of the target system
8. Scan completed successfully

Notes

- SYN Scan may require root/administrator privileges.
- Some firewalls or IDS/IPS may log SYN packets.
- Very effective for scanning large networks.
- Use `-Pn` if host is up but ICMP is blocked.

Tips

- Always start with SYN Scan.
- Combine with `-sV` for service detection.
- Use `-O` after SYN Scan for OS detection.
- Add `--reason` to see why ports are open/closed.

Stealthy scans lead to better situational awareness!

UDP Scan (-sU)

Scanning Connectionless Services – UDP Scan is used to discover UDP services that are listening on a target system. Unlike TCP, UDP does not establish a connection.

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What Is UDP?

- UDP (User Datagram Protocol) is connectionless.
- No handshake (like TCP).
- Faster but less reliable.
- Used by many important services.

Command

```
sudo nmap -sU 192.168.1.10
```

Run with sudo/administrator privileges. UDP scans are slower. Be patient!

How UDP Scan Works (Diagram)

Nmap sends a UDP probe (request) to the target (192.168.1.10). A response means the service is open. No response means filtered / closed.

Common UDP Services

Port	Service	Description
53/udp	DNS	Domain Name System
67/udp	DHCP (Server)	IP Address Assignment
68/udp	DHCP (Client)	IP Address Request
69/udp	TFTP	Trivial File Transfer
161/udp	SNMP	Simple Network Management
123/udp	NTP	Network Time Protocol
137/udp	NetBIOS Name	NetBIOS Name Service
138/udp	NetBIOS Datagram	NetBIOS Datagram Service
514/udp	Syslog	System Logging Service

TCP vs UDP Scan

TCP Scan	UDP Scan
Connection-oriented	Connectionless
Reliable	Less reliable
3-way handshake	No handshake
Easier to scan	Slower to scan

When to Use UDP Scan?

- Finding DNS servers

- Discovering SNMP devices
- Identifying DHCP servers
- Network enumeration in internal networks
- IoT and printer discovery

Real-World Example

A SOC analyst wants to find out if the DNS service is running on a server. UDP scan reveals that port 53/udp is open.

Kali Linux Output (Deep Explanation)

```
(kali@kali)-[~]
$ sudo nmap -sU 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 13:05 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).
Not shown: 997 closed udp ports (port-unreach)
PORT      STATE      SERVICE VERSION
53/udp    open|filtered domain  (DNS Server)
67/udp    open|filtered dhcps  (DHCP Server)
68/udp    open|filtered dhcpc  (DHCP Client)
123/udp   closed      ntp
161/udp   open|filtered snmp    SNMPv1 server (public)
514/udp   open|filtered syslog  syslogd 1.5.1
MAC Address: 00:0C:29:1A:2B:3C (VMware)

Nmap done: 1 IP address (1 host up) scanned in 15.62 seconds
```

1. Command used: UDP Scan on target
2. Nmap version and scan start time
3. Target IP that is being scanned
4. Host is up and response time
5. Closed UDP ports return “port-unreach”
6. Open or filtered UDP ports list (cannot differentiate without response)
7. Closed UDP port (explicitly closed)
8. Service and version detection (if available)
9. MAC address of the target system
10. Scan completed successfully

Notes

- UDP scans are slower than TCP scans.
- Firewalls may drop UDP packets silently.
- Use `-Pn` if host is up but not responding to ping.
- Open|Filtered means Nmap can't confirm if open.

Tips

- Start UDP scan with top 1000 ports.
- Use `--top-ports 1000` for faster results.
- Combine with `-sV` to detect versions.
- Use in authorized environments only.

Key Takeaway

UDP services are critical in many networks. Scanning them helps uncover hidden services!

Aggressive Scan (-A)

All-in-One Powerful Scan – Aggressive Scan enables multiple Nmap features at once to gather as much information about the target as possible.

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What Is -A Scan?

Aggressive Scan (-A) is Nmap's most comprehensive scan option. It combines multiple scanning techniques to collect detailed information in a single command.

Command

```
sudo nmap -A 192.168.1.10
```

Run with sudo/administrator privileges for best accuracy.

What Does -A Include?

Option	Description
-sV	Service Version Detection
-O	OS Detection
--script default	NSE Script Scan (Default Scripts)
--traceroute	Traceroute (if supported)
--reason	Reason for port state
--open	Show only open ports

How -A Scan Works (Diagram)

- **Service Detection** – Detects service versions running on open ports.
- **OS Detection** – Guess the operating system using TCP/IP fingerprinting.
- **NSE Scripts** – Runs default NSE scripts to gather more info.
- **Traceroute** – Discovers the path to the target (if network allows).
- **Reason & Open** – Shows reason for state and lists only open ports.

When to Use?

- Full network assessment
- Internal SOC investigations
- Incident response and threat hunting
- Before deeper vulnerability analysis
- When you need maximum information

Real-World Example

A SOC analyst investigates a suspicious server. Using -A scan, Nmap provides:

- Open ports
- Services and versions
- Operating system details
- Possible vulnerabilities (via scripts)
- Network path to the target

This helps the analyst quickly understand the risk and plan next steps.

Note

Aggressive scan can be noisy and takes more time. Use only in authorized environments.

Kali Linux Output (Deep Explanation)

```
(kali@kali)-[~]
$ sudo nmap -A 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 14:22 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
443/tcp   open  https    nginx 1.18.0 (Ubuntu)
3306/tcp  open  mysql    MySQL 5.7.33-0ubuntu0.18.04.1
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Device type: general purpose
Running (JUST GUESSING): Linux 4.X|5.X (90%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
Aggressive OS guesses: Linux 4.15 - 5.8 (90%), Linux 5.0 - 5.4 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   0.78 ms  192.168.1.10

OS and Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.45 seconds
```

1. Command used: Aggressive Scan (-A)
2. Nmap version and scan start time
3. Target IP scan result
4. Host is up with response time
5. Closed ports filtered by RST
6. Open ports with service & version info
7. OS detection (best guess)
8. OS accuracy / confidence
9. Traceroute path to the target
10. Scan summary and completion time

Tips

- Use -A for full enumeration.
- Requires more time.
- Use in authorized environments only.
- Combine with other scans for best results.

NSE Scripts

Nmap Scripting Engine

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What Is NSE?

Nmap Scripting Engine (NSE) allows Nmap to automate a wide range of tasks using scripts written in the Lua programming language. NSE makes Nmap extremely powerful for enumeration, discovery and vulnerability assessment.

NSE Workflow

Nmap (runs the scan) → NSE Script (executes the selected script) → Target (script interacts with the target) → Results (returns useful information).

Common NSE Script Categories

Category	Description
vuln	Detects known vulnerabilities
discovery	Discovers information about services
auth	Performs authentication related scripts
brute	Performs brute force attacks
malware	Detects malware, backdoors, C2
smb	Enumerates SMB/Windows related info
http	Enumerates HTTP / Web information
default	Default scripts (safe & basic)

Useful NSE Commands

```
nmap --script vuln 192.168.1.10
```

Check target for known vulnerabilities.

```
nmap --script smb-os-discovery 192.168.1.10
```

Discover OS, computer name, workgroup via SMB.

```
nmap --script http-title 192.168.1.10
```

Retrieve the title of the web page.

Real-World Example

A SOC analyst is investigating a suspicious Windows server (192.168.1.10) inside the network. Using NSE scripts, the analyst gathers important information quickly:

- Hostname and OS
- SMB details
- Web server information
- Potential vulnerabilities

Kali Linux Output (Deep Explanation)

```
(kali@kali)-[~]  
$ nmap --script vuln,smb-os-discovery,http-title 192.168.1.10
```

```

Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 16:18 IST
Nmap scan report for 192.168.1.10
Host is up (0.00078s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
| http-title: Welcome to Apache2 Ubuntu Default Page: It works
445/tcp    open  microsoft-ds?
| smb-os-discovery:
|   OS: Windows 10 Pro 19044 (Windows 10 Pro 6.3)
|   Computer name: WIN10-PRO
|   NetBIOS computer name: WIN10-PRO\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2024-05-24T10:48:33-07:00
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge response: supported
|_  smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

NSE: Script Post-scanning.
NSE: Starting runlevel 1 (of 2) scan.
NSE: Starting runlevel 2 (of 2) scan.
NSE: Script reporting.
Nmap done: 1 IP address (1 host up) scanned in 22.43 seconds

```

Explanation (Line by Line)

1. Command used: NSE scripts (vuln, smb-os-discovery, http-title)
2. Target scanned: 192.168.1.10
3. Target is up and responding
4. Open ports with services
5. HTTP title retrieved by script
6. SMB OS discovery results
7. SMB security information
8. Additional service / OS details
9. NSE script execution status
10. Scan summary and completion time

Notes

- NSE scripts enhance Nmap's capability beyond port scanning.
- Not all scripts are safe. Always use in authorized environments.
- Use `nmap --script-help=<script-name>` for more details.
- Scripts are written in Lua and stored in: `/usr/share/nmap/scripts/`

Key Takeaway

- NSE scripts = Automation + Information Gathering
- Helps SOC analysts save time and effort
- Provides deep insights about targets
- One of the most practical Nmap features

Finding Weaknesses with Nmap

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What Can Nmap Discover?

- Outdated software
- Misconfigured services
- Exposed ports
- Weak protocols
- Default credentials

Useful Commands

```
nmap --script vuln 192.168.1.10
nmap --script smb-vuln* 192.168.1.10
nmap --script http-vuln* 192.168.1.10
```

Real-World Example

A SOC analyst scans a web server and discovers an outdated Apache version which may have known vulnerabilities.

Kali Linux Output (Deep Explanation)

```
(kali@kali)-[~]
$ nmap --script vuln 192.168.1.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-05-24 16:35 IST
Nmap scan report for 192.168.1.10
Host is up (0.00065s latency).
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.5
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
445/tcp    open  microsoft-ds Windows 10 Pro 10.0
8080/tcp   open  http-proxy   nginx 1.18.0
|_ http-vuln-cve2021-41773:
|   VULNERABLE:
|   Apache 2.4.41 is vulnerable to Path Traversal (CVE-2021-41773)
|   Severity: HIGH
|   CVSS Score: 7.5 (High)
|   References: https://nvd.nist.gov/vuln/detail/CVE-2021-41773
|
|_ smb-vuln-ms17-010:
|   VULNERABLE: MS17-010 SMB Remote Code Execution
|   State: VULNERABLE
|_   Risk: HIGH

Nmap done: 1 IP address (1 host up) scanned in 38.62 seconds
```

Line-by-Line Explanation

1. Command used: vuln script
2. Target IP: 192.168.1.10
3. Open ports discovered
4. Service versions detected
5. http-vuln script found Apache Path Traversal vulnerability (CVE-2021-41773)
6. smb-vuln script found MS17-010 vulnerability in SMB service
7. Scan completed successfully

SOC Connection

- Used during Threat Hunting

- Incident Response
- Asset Discovery
- Compliance & Security Audits

Important Notes

- Nmap identifies possible issues. It does NOT exploit anything.
- Always verify findings manually.
- Use only in authorized labs or permitted environments.

Key Takeaway

- Nmap vulnerability scripts save analysts huge amount of time.
- Helps identify weak or risky services before attackers do.
- One of the most powerful features for Blue Teamers.

Real Incident Investigation

Complete SOC Analyst Workflow with Nmap

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Scenario

SIEM Alert Received: “Suspicious internal server detected communicating with external IP.” The SOC analyst starts investigation using Nmap.

Investigation Workflow

1. Alert Received
2. Host Discovery
3. Port Scan
4. Service Detection
5. OS Detection
6. NSE Scripts
7. Risk Assessment
8. Document Findings

Each step builds on the previous one to give a complete picture of the target.

Commands Used

Step	Purpose	Command
1	Host Discovery	<code>nmap -sn 192.168.1.0/24</code>
2	Port Scan (Stealth)	<code>sudo nmap -sS 192.168.1.10</code>
3	Service Detection	<code>nmap -sV 192.168.1.10</code>
4	OS Detection	<code>sudo nmap -O 192.168.1.10</code>
5	NSE Vulnerability Scan	<code>nmap --script vuln 192.168.1.10</code>

Deep Explanation (Step by Step)

1. **Host Discovery:** Find live hosts in the network.
2. **Port Scan (SYN):** Find open ports using stealth scan.
3. **Service Detection (-sV):** Identify running services and versions.
4. **OS Detection (-O):** Detect the operating system of the target.
5. **NSE Scripts (vuln):** Check for known vulnerabilities.
6. **Risk Assessment:** Analyze results and determine potential risk.
7. **Documentation:** Document findings for reporting and response.

Analyst Tip

Start broad, then narrow down. Gather as much information as possible, then validate and correlate with other tools (SIEM, EDR, Logs).

Kali Linux Output (Complete Investigation Example)

```
(kali@kali)-[~]
$ nmap -sn 192.168.1.0/24
Nmap scan report for 192.168.1.10
Host is up (0.00055s latency).
Nmap scan report for 192.168.1.15
Host is up (0.00041s latency).
```

```

$ sudo nmap -sS 192.168.1.10
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
445/tcp    open  microsoft-ds
8080/tcp   open  http-proxy

$ nmap -sV 192.168.1.10
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http      Apache httpd 2.4.41 ((Ubuntu))
445/tcp    open  microsoft-ds Windows 10 Pro 10.0
8080/tcp   open  http-proxy nginx 1.18.0

$ sudo nmap -O 192.168.1.10
OS details: Microsoft Windows 10 Pro 10.0.19045 (Windows 10 Pro 2009 Update)
Network Distance: 1 hop

$ nmap --script vuln 192.168.1.10
|_ http-vuln-cve2021-41773: Apache 2.4.41 is vulnerable to Path Traversal (CVE-2021-41773)
|_ smb-vuln-ms17-010: VULNERABLE:
|   Remote Code Execution vulnerability in SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   Risk factor: HIGH
|_ References: https://nvd.nist.gov/vuln/detail/CVE-2017-0144

Nmap done: 1 IP address (1 host up) scanned in 45.32 seconds

```

Output Callouts

1. Host discovery found live hosts (192.168.1.10 and 192.168.1.15).
2. SYN scan found open ports: 22, 80, 445, 8080.
3. Service detection revealed versions of running services.
4. OS detection identified the target as Windows 10 Pro.
5. NSE vulnerability scan found: Apache Path Traversal (CVE-2021-41773) and SMBv1 Remote Code Execution (MS17-010).
6. Scan completed successfully in 45.32 seconds.

What We Learned

- Target is a Windows 10 Pro system.
- Web server (Apache 2.4.41) is vulnerable.
- SMBv1 is enabled and vulnerable.
- This host should be prioritized for patching and monitoring.

Nmap Quick Reference

Cheat Sheet + Blue Team Roadmap

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1. Most Important Commands

Scan	Purpose	Command
Host Discovery	Find live hosts	<code>nmap -sn 192.168.1.0/24</code>
Port Scan (TCP)	Scan for open TCP ports	<code>nmap -sS 192.168.1.10</code>
Version Detection	Identify service versions	<code>nmap -sV 192.168.1.10</code>
OS Detection	Detect operating system	<code>nmap -O 192.168.1.10</code>
Aggressive Scan	Enable OS, version, script, tracing & more	<code>nmap -A 192.168.1.10</code>
UDP Scan	Scan UDP ports	<code>nmap -sU 192.168.1.10</code>
NSE Scripts	Run Nmap scripts	<code>nmap --script vuln 192.168.1.10</code>
Save Results	Save output to file	<code>nmap -A 192.168.1.10 -oN result.txt</code>

2. SOC Analyst Nmap Workflow

1. **Discover Hosts** – Find live systems in the network.
2. **Find Open Ports** – Identify which ports are open.
3. **Identify Services** – Detect running services & versions.
4. **Detect OS** – Determine the operating system.
5. **Run NSE Scripts** – Check for vulnerabilities & gather info.
6. **Assess Risks** – Analyze exposure & potential impact.
7. **Document Findings** – Report & take action.

3. What's Next After Nmap?

- **Wireshark** – Packet analysis & deep network visibility
- **Burp Suite** – Web application security testing
- **Wazuh** – Threat detection & log analysis (SIEM)
- **Splunk** – SIEM & data analysis platform
- **Suricata** – Network IDS/IPS & threat detection
- **TheHive** – Incident response & case management
- **Threat Hunting** – Proactively find threats in your environment

Bonus: Nmap Tips

- Start slow and steady. Avoid noisy scans.
- Use `-A` only when you have permission.
- Combine Nmap with other tools.
- Always verify findings manually.
- Save results and keep evidence.
- In SOC, every detail matters!

“You cannot defend what you cannot see. Nmap gives defenders visibility.”

Your Blue Team Journey Roadmap

Learn Fundamentals → Practice Nmap → Understand Networks → Analyze & Investigate →
Hunt Threats → Protect & Respond → Keep Learning & Improving

Keep practicing, keep learning, stay curious!

STAY CURIOUS. STAY SHARP. STAY BLUE. You've got this!